

Reproduction of the Paper’s Table 6

Table 1: Paper, Table 6.

<i>Strategy</i>	<i>Vanilla FL</i>			<i>Global-DP</i>			<i>Metric-privacy</i>		
	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>
<i>FedAvg</i>	0.909	0.903	0.950	0.884	0.862	0.918	0.894	0.843	0.943
<i>FedAvgM</i>	0.884	0.768	0.932	0.762	0.554	0.607	0.823	0.608	0.631
<i>FedMedian</i>	0.932	0.899	0.946	0.875	0.827	0.913	0.895	0.843	0.943
<i>FedProx</i>	0.909	0.848	0.873	0.877	0.853	0.914	0.903	0.877	0.944
<i>FedOpt</i>	0.950	0.936	0.970	0.918	0.877	0.911	0.930	0.917	0.950
<i>FedYogi</i>	0.933	0.914	0.957	0.908	0.841	0.884	0.917	0.888	0.941

Table 2: Ours (mean $\pm$ std over 3 seeds).

<i>Strategy</i>	<i>Vanilla FL</i>			<i>Global-DP</i>			<i>Metric-privacy</i>		
	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>
<i>FedAvg</i>	0.951 $\pm$ 0.004	0.950 $\pm$ 0.004	0.951 $\pm$ 0.004	0.942 $\pm$ 0.004	0.942 $\pm$ 0.004	0.942 $\pm$ 0.004	0.945 $\pm$ 0.007	0.945 $\pm$ 0.007	0.945 $\pm$ 0.007

Table 3: Difference (ours – paper).

<i>Strategy</i>	<i>Vanilla FL</i>			<i>Global-DP</i>			<i>Metric-privacy</i>		
	Δ Acc.	Δ F1	Δ Prec.	Δ Acc.	Δ F1	Δ Prec.	Δ Acc.	Δ F1	Δ Prec.
<i>FedAvg</i>	+0.042	+0.047	+0.001	+0.058	+0.080	+0.024	+0.051	+0.102	+0.002

Table 4: Ours, macro-averaged F1/precision.

<i>Strategy</i>	<i>Vanilla FL</i>			<i>Global-DP</i>			<i>Metric-privacy</i>		
	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>
<i>FedAvg</i>	0.951 $\pm$ 0.004	0.936 $\pm$ 0.006	0.936 $\pm$ 0.027	0.942 $\pm$ 0.004	0.946 $\pm$ 0.007	0.955 $\pm$ 0.004	0.945 $\pm$ 0.007	0.941 $\pm$ 0.011	0.959 $\pm$ 0.004

Table 5: Previous runs, `results/reproduce_paper/` (homogeneous, noise 0.01, mean $\pm$ std over 5 seeds).

<i>Strategy</i>	<i>Vanilla FL</i>			<i>Global-DP</i>			<i>Metric-privacy</i>		
	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>
<i>FedAvg</i>	0.958 $\pm$ 0.012	0.958 $\pm$ 0.012	0.959 $\pm$ 0.011	0.942 $\pm$ 0.012	0.942 $\pm$ 0.012	0.943 $\pm$ 0.012	0.945 $\pm$ 0.009	0.945 $\pm$ 0.009	0.945 $\pm$ 0.009
<i>FedAvgM</i>	0.919 $\pm$ 0.017	0.918 $\pm$ 0.017	0.921 $\pm$ 0.016	0.777 $\pm$ 0.027	0.773 $\pm$ 0.029	0.773 $\pm$ 0.030	0.792 $\pm$ 0.020	0.787 $\pm$ 0.024	0.788 $\pm$ 0.026
<i>FedMedian</i>	0.947 $\pm$ 0.008	0.947 $\pm$ 0.007	0.947 $\pm$ 0.007	0.944 $\pm$ 0.006	0.944 $\pm$ 0.006	0.944 $\pm$ 0.006	0.944 $\pm$ 0.008	0.944 $\pm$ 0.007	0.945 $\pm$ 0.007
<i>FedProx</i>	0.550 $\pm$ 0.014	0.508 $\pm$ 0.015	0.474 $\pm$ 0.017	0.549 $\pm$ 0.018	0.506 $\pm$ 0.027	0.499 $\pm$ 0.076	0.496 $\pm$ 0.001	0.329 $\pm$ 0.002	0.317 $\pm$ 0.161
<i>FedOpt</i>	0.411 $\pm$ 0.077	0.272 $\pm$ 0.075	0.210 $\pm$ 0.078	0.495 $\pm$ 0.000	0.328 $\pm$ 0.000	0.245 $\pm$ 0.000	—	—	—
<i>FedYogi</i>	0.951 $\pm$ 0.016	0.953 $\pm$ 0.014	0.956 $\pm$ 0.012	0.947 $\pm$ 0.012	0.947 $\pm$ 0.012	0.947 $\pm$ 0.012	0.952 $\pm$ 0.010	0.952 $\pm$ 0.010	0.953 $\pm$ 0.009

Table 6: Previous runs, macro-averaged F1/precision.

<i>Strategy</i>	<i>Vanilla FL</i>			<i>Global-DP</i>			<i>Metric-privacy</i>		
	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>	<i>Acc.</i>	<i>F1</i>	<i>Prec.</i>
<i>FedAvg</i>	0.958 $\pm$ 0.012	0.939 $\pm$ 0.040	0.933 $\pm$ 0.059	0.942 $\pm$ 0.012	0.936 $\pm$ 0.022	0.947 $\pm$ 0.026	0.945 $\pm$ 0.009	0.941 $\pm$ 0.025	0.957 $\pm$ 0.011
<i>FedAvgM</i>	0.919 $\pm$ 0.017	0.863 $\pm$ 0.078	0.937 $\pm$ 0.015	0.777 $\pm$ 0.027	0.603 $\pm$ 0.097	0.621 $\pm$ 0.134	0.792 $\pm$ 0.020	0.661 $\pm$ 0.124	0.684 $\pm$ 0.151
<i>FedMedian</i>	0.947 $\pm$ 0.008	0.945 $\pm$ 0.012	0.947 $\pm$ 0.012	0.944 $\pm$ 0.006	0.940 $\pm$ 0.014	0.954 $\pm$ 0.006	0.944 $\pm$ 0.008	0.942 $\pm$ 0.010	0.947 $\pm$ 0.020
<i>FedProx</i>	0.550 $\pm$ 0.014	0.289 $\pm$ 0.009	0.269 $\pm$ 0.009	0.549 $\pm$ 0.018	0.289 $\pm$ 0.019	0.318 $\pm$ 0.119	0.496 $\pm$ 0.001	0.166 $\pm$ 0.001	0.174 $\pm$ 0.112
<i>FedOpt</i>	0.411 $\pm$ 0.077	0.157 $\pm$ 0.024	0.120 $\pm$ 0.035	0.495 $\pm$ 0.000	0.166 $\pm$ 0.000	0.124 $\pm$ 0.000	—	—	—
<i>FedYogi</i>	0.951 $\pm$ 0.016	0.911 $\pm$ 0.076	0.907 $\pm$ 0.079	0.947 $\pm$ 0.012	0.927 $\pm$ 0.028	0.948 $\pm$ 0.019	0.952 $\pm$ 0.010	0.952 $\pm$ 0.014	0.961 $\pm$ 0.015

Table 7: Per-seed final-model test metrics.

Mechanism	Strategy	Seed	Weighted avg.			Macro avg.			Log-loss
			Acc.	F1	Prec.	F1	Prec.	Rec.	
Vanilla FL	FedAvg	42	0.9547	0.9546	0.9553	0.9287	0.9126	0.9527	0.1992
Vanilla FL	FedAvg	43	0.9469	0.9469	0.9471	0.9410	0.9295	0.9548	0.2977
Vanilla FL	FedAvg	44	0.9500	0.9497	0.9505	0.9372	0.9656	0.9130	0.2901
Global-DP	FedAvg	42	0.9391	0.9389	0.9391	0.9491	0.9542	0.9445	0.2432
Global-DP	FedAvg	43	0.9406	0.9407	0.9410	0.9502	0.9514	0.9492	0.2801
Global-DP	FedAvg	44	0.9469	0.9468	0.9469	0.9380	0.9602	0.9187	0.2650
Metric-privacy	FedAvg	42	0.9500	0.9497	0.9498	0.9335	0.9573	0.9130	0.2287
Metric-privacy	FedAvg	43	0.9375	0.9375	0.9376	0.9533	0.9568	0.9501	0.2861
Metric-privacy	FedAvg	44	0.9484	0.9481	0.9489	0.9367	0.9642	0.9131	0.2843